Chronic Disease Risk Analysis Dashboard

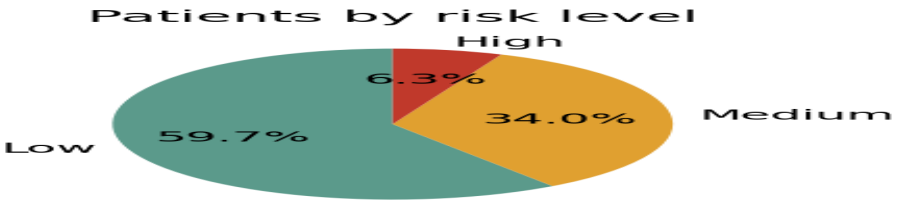
Heart disease, diabetes and stroke across 600 patients • Synthetic sample data • Educational use only

1. Clinical Risk Analysis

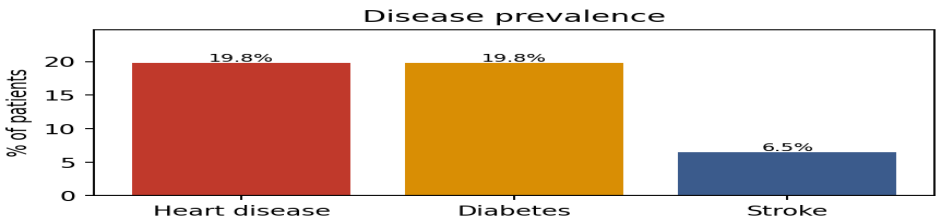
Summary indicators, disease prevalence, patient risk distribution and the highest-risk patient records.

600	6.3%	26.6	116	192	99
Patients	High risk	Average BMI	Avg systolic BP	Avg cholesterol	Avg glucose

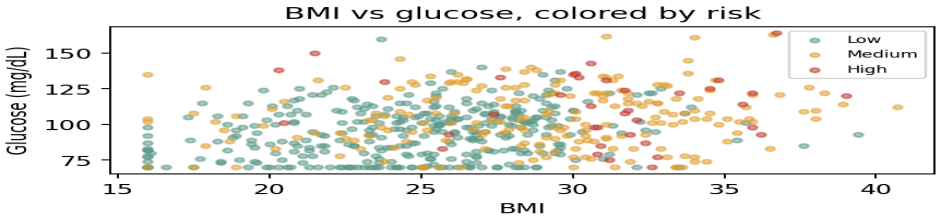
Patients by risk level



Disease prevalence (%)



BMI vs glucose, colored by risk



Highest-risk patients

See highest-risk patient table below.

Highest-risk patients (top 8 by risk score, then age)

Patient	Age	Blood pressure	Cholesterol	Glucose	Risk score / level
P1273	79	127/89	272	138	6 / High
P1126	69	121/69	234	133	6 / High
P1212	48	147/92	197	135	6 / High
P1349	70	148/90	182	114	5 / High
P1365	70	142/81	190	98	5 / High
P1233	66	121/65	240	122	5 / High
P1148	63	137/84	243	110	5 / High
P1369	60	147/94	288	98	5 / High

Risk score is a dataset-defined indicator, not a clinical diagnosis. Sample dataset for learning; not for clinical use.

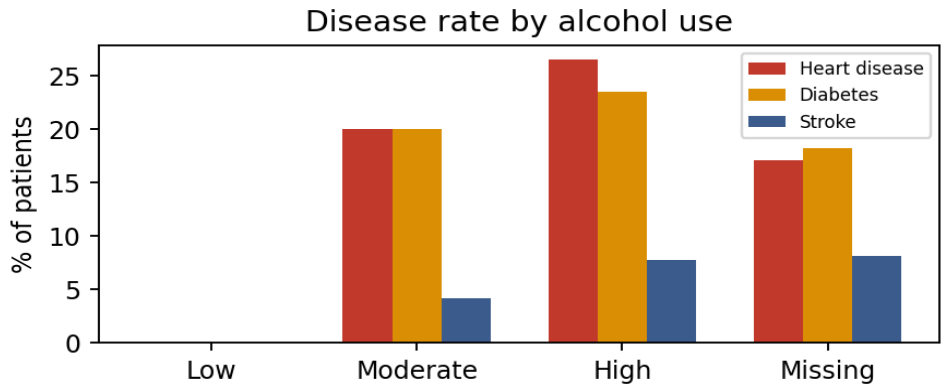
2. Lifestyle Analysis

Explores smoking, exercise, alcohol-use categories, glucose by age group and disease rates associated with listed symptoms.

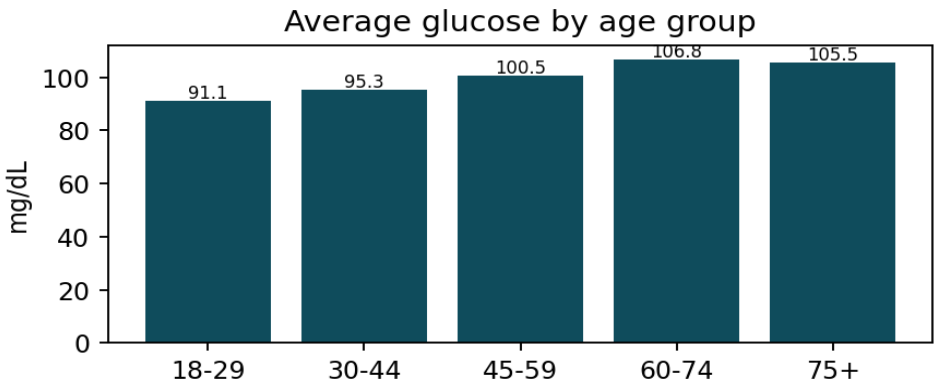
Heart disease rate (%) by smoking × exercise

Exercise level	Never	Former	Current
Low	24.5% (n=139)	21.3% (n=47)	36.5% (n=63)
Medium	15.0% (n=140)	7.7% (n=39)	18.9% (n=53)
High	16.1% (n=62)	3.7% (n=27)	23.3% (n=30)

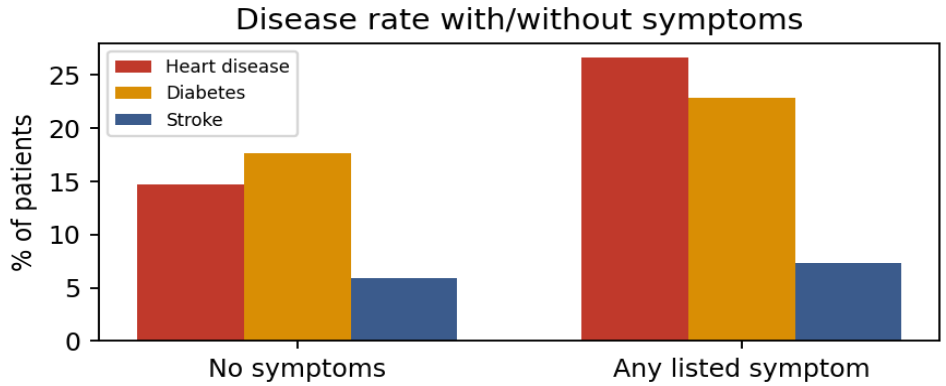
Disease rate (%) by alcohol use



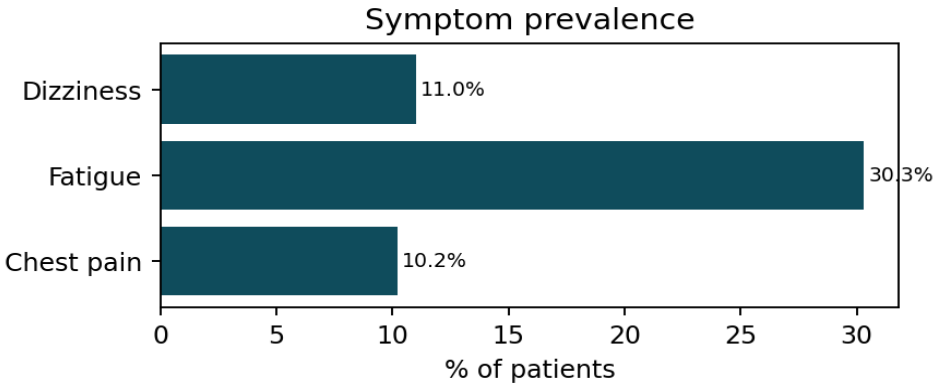
Average glucose by age group (mg/dL)



Heart disease rate with/without listed symptoms



Symptom prevalence reference

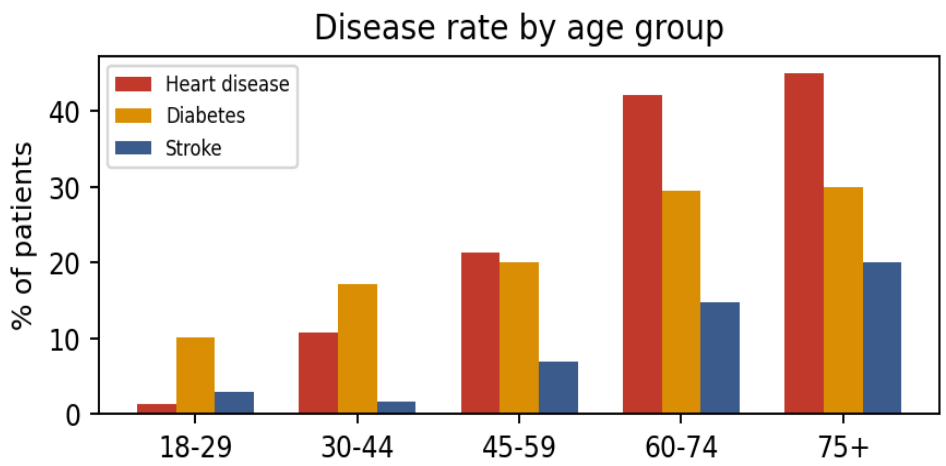


Alcohol-use categories with missing values are shown as “Missing”. “Any listed symptom” means at least one of chest pain, fatigue or dizziness is marked Yes.

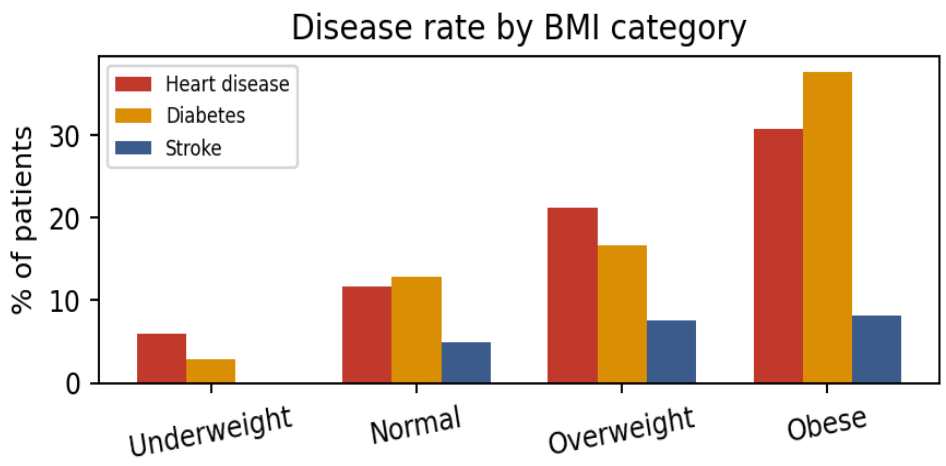
3. Population Analysis

Compares disease rates across age and BMI groups, distribution of exercise levels and symptom prevalence.

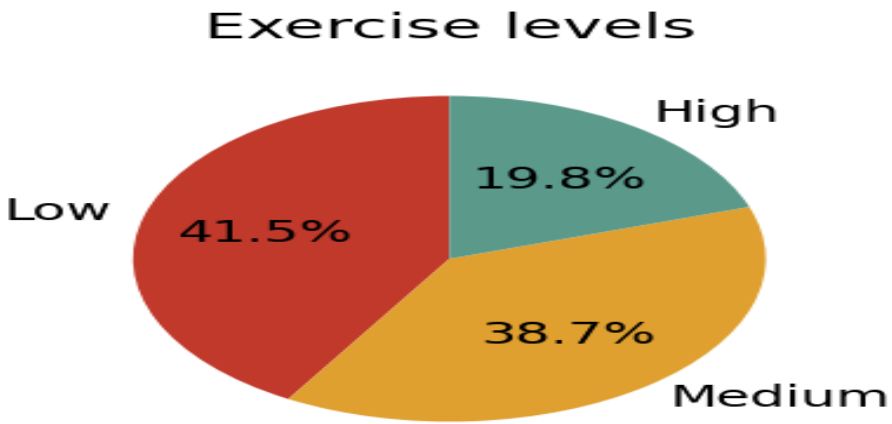
Disease rate (%) by age group



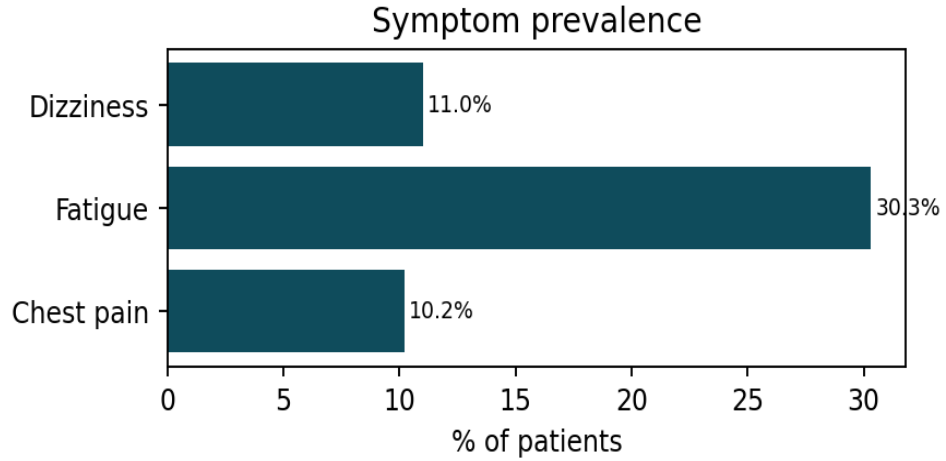
Disease rate (%) by BMI category



Exercise levels



Symptom prevalence (%)



Interpretation note: all records are synthetic sample data. Relationships shown are descriptive and should not be interpreted as medical advice or causal evidence.

Health360: Data Visualization Guide (Tableau)

Project: Health360 – Comprehensive Healthcare Risk & Disease Analytics
Program: Data Analytics with Tableau (Skillwallet)

1. Calculated Fields Used

Age Group

```
IF [Age] < 18 THEN "0-17"
ELSEIF [Age] < 30 THEN "18-29"
ELSEIF [Age] < 45 THEN "30-44"
ELSEIF [Age] < 60 THEN "45-59"
ELSE "60+" END
```

BMI Category

```
IF [BMI] < 18.5 THEN "Underweight"
ELSEIF [BMI] < 25 THEN "Normal"
ELSEIF [BMI] < 30 THEN "Overweight"
ELSE "Obese" END
```

Any Disease

```
IF [Heart Disease]=1 OR [Diabetes]=1 OR [Stroke]=1 THEN 1 ELSE 0 END
```

Risk Score

```
(IF [Smoking]=1 THEN 1 ELSE 0 END)
+ (IF [Alcohol]=1 THEN 1 ELSE 0 END)
+ (IF [BMI]>=30 THEN 1 ELSE 0 END)
+ (IF [Cholesterol]>=240 THEN 1 ELSE 0 END)
+ (IF [Glucose]>=126 THEN 1 ELSE 0 END)
+ (IF [Blood Pressure]>=140 THEN 1 ELSE 0 END)
```

Risk Level

```
IF [Risk Score]>=4 THEN "High"
ELSEIF [Risk Score]>=2 THEN "Medium"
ELSE "Low" END
```

Measures

```
Total Patients = COUNTD([Patient ID])
Diseased Patients = SUM([Any Disease])
Prevalence % = SUM([Any Disease]) / COUNTD([Patient ID])
```

Format *Prevalence %* as Percentage with 1 decimal. Adjust 1/0 conditions to Yes/No if the dataset uses text values.

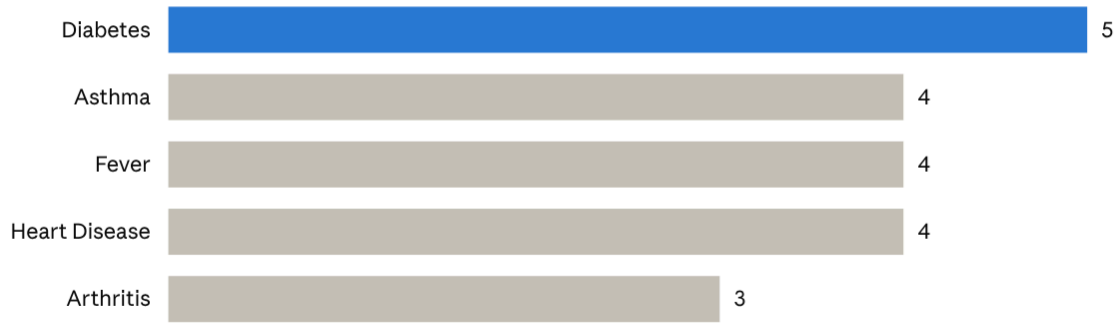
2. Sheets and Charts

#	Sheet	Chart type	Fields
1	KPI Cards	Text	Total Patients, Prevalence %, High-Risk count, Avg BMI
2	Disease Distribution	Bar	Heart Disease, Diabetes, Stroke
3	Prevalence by Age Group	Bar	Age Group, Prevalence %
4	Disease by BMI Category	Stacked bar	BMI Category, Diseased Patients
5	Risk Level Split	Pie / Donut	Risk Level, Total Patients
6	Smoking and Alcohol vs Disease	Side-by-side bar	Smoking, Alcohol, Prevalence %
7	Exercise Level vs Disease	Bar	Exercise Level, Prevalence %
8	Heatmap	Heatmap	Age Group, BMI Category, Prevalence %
9	Symptom Prevalence	Horizontal bar	Chest Pain, Fatigue, Dizziness
10	BMI vs Glucose	Scatter	BMI, Glucose, Any Disease, Cholesterol

3. Insights

Diabetes is the most common diagnosis: 5 of 20 patients

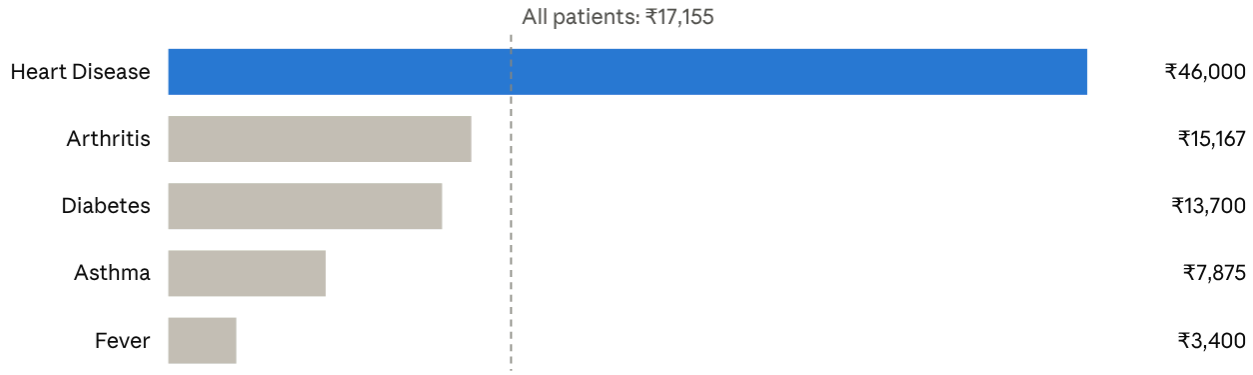
20 patients across 5 diseases



healthcare_360_sample_dataset.xlsx · 20 patients, Jan–Mar 2026

Heart disease costs ₹46,000 per patient, 3x the next-highest disease

Average treatment cost per patient (INR), 20 patients



healthcare_360_sample_dataset.xlsx · Treatment_Cost_INR, 20 patients

Heart disease patients also stay longest, 9.5 days on average against 2.0 for fever.

HEALTHCARE ANALYTICS CASE STUDY

Early Detection & Risk Intelligence for Chronic Diseases

Executive Summary

Chronic conditions such as Heart Disease, Diabetes, and Stroke pose severe challenges to global healthcare systems. Delayed diagnosis amplifies patient complications and inflates medical expenditures. This project delivers a production-ready **Tableau Health Analytics Suite** linking demographic data, clinical vital signs, and behavioral indicators. By translating complex healthcare data into intuitive visual workflows, this solution empowers clinical teams and public health leaders to transition from reactive treatment to proactive risk prevention.

1. THE PROBLEM & STRATEGIC CONTEXT

Modern healthcare systems capture immense volumes of clinical and lifestyle data. However, critical patient indicators—ranging from blood pressure trends and fasting glucose to routine exercise habits—frequently remain isolated in disconnected data silos and spreadsheets. Manual analysis makes it nearly impossible for care managers to detect multi-variable risk patterns early, leading to delayed medical interventions and preventable emergency admissions.

2. PERSONA-DRIVEN CLINICAL SCENARIOS

Dr. Michael Carter

CLINICAL RISK ASSESSMENT OFFICER

Challenge: Difficulty spotting high-risk patients quickly due to manual record processing, leading to delayed interventions.

Solution & Workflow: Uses Tableau early-warning scorecards and scatter plots combining BMI, blood pressure, cholesterol, and glucose.

Outcome: Swift identification of critical cohorts for immediate clinical follow-up, reducing complications.

Emily Rodriguez

HEALTHCARE DATA ANALYST

Challenge: Identifying how lifestyle choices compound health risks across heterogeneous patient populations.

Solution & Workflow: Develops correlation matrices and lifestyle heatmaps linking exercise, smoking, and alcohol intake to disease prevalence.

Outcome: Informs clinical teams with data-backed evidence to design targeted preventive care plans.

Dr. Priya Nair

PUBLIC HEALTH RESEARCHER

Challenge: Analyzing population-wide disease trends and demographic vulnerability without clear visual aggregation tools.

Solution & Workflow: Analyzes age-group distributions, BMI classifications, and symptom frequency maps (fatigue, chest pain, dizziness).

Outcome: Guides strategic allocation of public health resources, community screening drives, and preventive healthcare policies.

3. ANALYZED DATA ARCHITECTURE

Category	Attributes & Indicators	Analytical Purpose
Demographics	Age, Age Group Categories	Cohort stratification and vulnerable population mapping
Clinical Vitals	BMI, Blood Pressure, Cholesterol, Glucose	Baseline risk evaluation and metabolic health scoring
Lifestyle Factors	Smoking Habits, Alcohol Intake, Exercise Frequency	Behavioral correlation and compound risk modeling
Symptomatic Profiles	Fatigue Conditions, Chest Pain, Dizziness Patterns	Early warning indicator tracking and symptom severity analysis

4. TABLEAU VISUAL ARCHITECTURE

Dashboard 1: Executive Clinical Risk Matrix

Integrates dynamic KPI cards (High-Risk Count, Average BMI, Glucose Alerts) with interactive scatter plots to highlight immediate intervention targets for clinical staff.

Dashboard 2: Behavioral Risk Factor Heatmap

Maps lifestyle habits against chronic disease rates, enabling analysts to isolate compound risk vectors driven by smoking, inactivity, and alcohol use.

Dashboard 3: Population Demographics & Symptom Profiler

Displays stacked distribution charts of disease prevalence across age cohorts and BMI categories to assist public health resource planning.

5. PROJECT IMPACT & OUTCOMES

Proactive Care Shift

Transitions clinical operations from reactive treatment to timely, early-stage intervention.

360° Risk Insights

Connects daily lifestyle habits directly with clinical diagnostic vital indicators.

SKILLWALLET PORTFOLIO PROJECT STORY

Early Detection & Risk Intelligence for Chronic Diseases

Transforming Multi-Dimensional Health Data into Actionable Tableau Visual Intelligence

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